IT Infrastructure Test

PolyWorks Mexico

Instructions: You can answer your test handwriting or digitally, feel free to add screenshots if you need it.

1. Write a script in PowerShell to copy the folder c:/sourcefolder and its content to this location d:/destination. The date and time need to be inserted in the folder's name. The script is going to be executed daily indefinitely but ensure there are always just 10 folders in the destination location.

   Output example:

   D:/destination/sourcefolder_10202022-1200

```
# Q1_backup_rotation.ps1
$source      = "C:\sourcefolder"
$destination = "D:\destination"
$maxFolders  = 10
# Nombre con formato: sourcefolder_MMddyyyy-HHmm
$timestamp  = Get-Date -Format "MMddyyyy-HHmm"
$folderName = "sourcefolder_$timestamp"
$targetPath = Join-Path $destination $folderName
# Copiar carpeta completa
Copy-Item -Path $source -Destination $targetPath -Recurse -Force
Write-Host "Copia creada: $targetPath"
# Obtener todas las carpetas en destino, ordenadas de más antigua a más nueva
$folders = Get-ChildItem -Path $destination -Directory |
```

```
        Sort-Object CreationTime
# Eliminar las más antiguas si se superan los 10 folders
while ($folders.Count -gt $maxFolders) {
   $oldest = $folders[0]
   Remove-Item -Path $oldest.FullName -Recurse -Force
   Write-Host "Eliminada carpeta antigua: $($oldest.FullName)"
   # Re-evaluar la lista
   $folders = Get-ChildItem -Path $destination -Directory |
          Sort-Object CreationTime
}
```

PROGRAMARLO DIARIAMENTE

```
$action  = New-ScheduledTaskAction -Execute "powershell.exe" `
        -Argument "-NonInteractive -File C:\scripts\Q1_backup_rotation.ps1"
$trigger = New-ScheduledTaskTrigger -Daily -At "12:00AM"
Register-ScheduledTask -TaskName "BackupRotation" -Action $action `
                -Trigger $trigger -RunLevel Highest -Force
```

2. Write a script in PowerShell to get the following system information:

   CPU usage

   Memory usage

   Disk usage

   Save each output string into an array and print all the information to a file.

```
# Q2_system_info.ps1
$outputFile = "C:\logs\system_info_$(Get-Date -Format 'yyyyMMdd-HHmm').txt"
$info = @()   # Array donde se acumulan los strings
```

```
# --- CPU Usage ---
$cpu = Get-CimInstance Win32_Processor |
    Measure-Object -Property LoadPercentage -Average
$info += "CPU Usage:    $([math]::Round($cpu.Average, 2)) %"
# --- Memory Usage ---
$os     = Get-CimInstance Win32_OperatingSystem
$totalMB = [math]::Round($os.TotalVisibleMemorySize / 1KB, 2)
$freeMB  = [math]::Round($os.FreePhysicalMemory    / 1KB, 2)
$usedMB  = $totalMB - $freeMB
$memPct  = [math]::Round(($usedMB / $totalMB) * 100, 2)
$info += "Memory Usage: $usedMB MB used of $totalMB MB ($memPct %)"
# --- Disk Usage (todos los volúmenes fijos) ---
$disks = Get-CimInstance Win32_LogicalDisk -Filter "DriveType=3"
foreach ($d in $disks) {
   $totalGB = [math]::Round($d.Size       / 1GB, 2)
   $freeGB  = [math]::Round($d.FreeSpace   / 1GB, 2)
   $usedGB  = [math]::Round(($d.Size - $d.FreeSpace) / 1GB, 2)
   $pct     = [math]::Round(($usedGB / $totalGB) * 100, 2)
   $info   += "Disk $($d.DeviceID)  $usedGB GB used of $totalGB GB ($pct %)"
}
# --- Imprimir en consola y guardar en archivo ---
$info | ForEach-Object { Write-Host $_ }
New-Item -ItemType Directory -Path (Split-Path $outputFile) -Force | Out-Null
$info | Out-File -FilePath $outputFile -Encoding UTF8
Write-Host "`nResultados guardados en: $outputFile"
```

3. Given the following network address 172.17.0.0, create 30 subnets and each network must support 1800 hosts. Specify which subnet mask you are going to use and describe why. You can answer handwriting no specific tool or CLI is needed.

N de bits de host: 11 = 2046 host

Mascara de subred: 32 – 11 = 21 bits de red → /21 → 255.255.248.0

Numero de dirección de red	Dirección de red	Primera dirección de host	Ultima dirección de host	Broadcast
1	172.17.0.0 /21	172.17.0.1	172.17.7.254	172.17.7.255
2	172.17.8.0 /21	172.17.8.1	172.17.15.254	172.17.15.255
3	172.17.16.0 /21	172.17.16.1	172.17.23.254	172.17.23.255
.....	.....	.....	.....	.....
30	172.17.232.0 /21	172.17.232.1	172.17.239.254	172.17.239.255

4. Using PowerShell and commands of any virtualization engine (VMware, Hyper-V, VirtualBox) write a script that executes the following actions:

   • Create a virtual machine with the specs below (assume that the OS is going to be installed later)

   o Ram: 8GB

   o CPU: 4cpus

   o Storage: 150GB

   o Location: (default or random location)

   Note: You can use random values if needed.

   The virtual machine is going to be up only during working hours

   Write a script and actions to schedule the VM to start at 9:00 am and shut down at 23:00 hrs.

CREAR LA VM

```
# Q4_create_vm.ps1
$VMName   = "PolyWorks-VM-$(Get-Random -Maximum 9999)"
$RAMBytes = 8GB
$CPUCount = 4
$DiskGB   = 150
$VMPath   = $null    # null = default de Hyper-V
# Crear la VM sin OS (GenerationHide = disco vacío, se instalará después)
New-VM -Name $VMName -MemoryStartupBytes $RAMBytes -Generation 2 `
    -Path ($VMPath ?? (Get-VMHost).VirtualMachinePath)
# Configurar CPUs
Set-VMProcessor -VMName $VMName -Count $CPUCount
# Crear y conectar disco de 150 GB
$vhdPath = Join-Path (Get-VMHost).VirtualHardDiskPath "$VMName-OS.vhdx"
New-VHD -Path $vhdPath -SizeBytes ($DiskGB * 1GB) -Dynamic
Add-VMHardDiskDrive -VMName $VMName -Path $vhdPath
Write-Host "VM '$VMName' creada. Recuerda montar una ISO para instalar el SO."
```

SCRIPT DE ARRANQUE

```
# Q4_start_vm.ps1
$VMName = "PolyWorks-VM-XXXX"
Start-VM -Name $VMName
Write-Host "VM iniciada: $(Get-Date)"
```

SCRIPT DE APAGADO

```
# Q4_stop_vm.ps1
$VMName = "PolyWorks-VM-XXXX"
Stop-VM -Name $VMName -Force
```

```
Write-Host "VM detenida: $(Get-Date)"
```

REGISTRAR LAS TAREAS PROGRAMADAS

```
# Inicio 9:00 am
$startAction  = New-ScheduledTaskAction -Execute "powershell.exe" `
            -Argument "-NonInteractive -File C:\scripts\Q4_start_vm.ps1"
$startTrigger = New-ScheduledTaskTrigger -Daily -At "09:00AM"
Register-ScheduledTask -TaskName "VM_Start" -Action $startAction `
                -Trigger $startTrigger -RunLevel Highest -Force

# Apagado 23:00
$stopAction  = New-ScheduledTaskAction -Execute "powershell.exe" `
           -Argument "-NonInteractive -File C:\scripts\Q4_stop_vm.ps1"
$stopTrigger = New-ScheduledTaskTrigger -Daily -At "11:00PM"
Register-ScheduledTask -TaskName "VM_Stop" -Action $stopAction `
                -Trigger $stopTrigger -RunLevel Highest -Force
```

5. Sketch a basic structure to deploy an Active Directory for a small company, considering the objects, groups, users, and some departments. It’s an open sketch you can use any type of representation for your answer.

